

☒ Easy Level (10 Questions)

Q1. What is the main function of a VPN?

- A) Speed up internet
- B) Provide secure connection over the internet
- C) Host websites
- D) Monitor traffic

Answer: B

Q2. Which protocol is **obsolete** and considered insecure?

- A) L2TP
- B) OpenVPN
- C) PPTP
- D) IPSec

Answer: C

Q3. Which protocol is **open-source** and widely used for secure connections?

- A) PPTP
- B) OpenVPN
- C) GRE
- D) FTP

Answer: B

Q4. A **Remote Access VPN** is typically used by:

- A) Data centers
- B) Government labs
- C) Individual users to access a private network
- D) Branch offices

Answer: C

Q5. Which of the following does a VPN help hide?

- A) Server uptime
- B) Physical network topology
- C) User's real IP address
- D) Password hashes

Answer: C

Q6. IPSec operates at which layer?

- A) Application
- B) Transport
- C) Network
- D) Session

Answer: C

Q7. Which protocol uses port **443** and is good at bypassing firewalls?

- A) L2TP
- B) SSL VPN
- C) GRE
- D) IPSec

Answer: B

Q8. Trusted VPNs are commonly used with which transport mechanism?

- A) MPLS
- B) HTTP
- C) SMTP
- D) ICMP

Answer: A

Q9. WireGuard is known for:

- A) Legacy encryption
- B) Slow performance
- C) Simplicity and high speed
- D) GUI-based control only

Answer: C

Q10. The key feature of a **Secure VPN** is:

- A) Port forwarding
- B) Strong encryption and tunneling
- C) Speed testing
- D) User monitoring

Answer: B

☒ Next, I'll send **20 Medium-Level Questions**, then 10 Hard. Ready?

You said:

yes

ChatGPT said:

Perfect! Here's the continuation with:

📌 VPN – 20 Medium-Level MCQs

Q11. Which of the following VPN protocols uses UDP port 1194 by default?

- A) PPTP
- B) L2TP
- C) OpenVPN
- D) SSL VPN

Answer: C

📌 *OpenVPN by default uses UDP port 1194 unless otherwise configured.*

Q12. In IPSec, which mode **encrypts the entire IP packet**?

- A) Transport Mode
- B) Tunnel Mode
- C) NAT-T Mode
- D) Secure Mode

Answer: B

Q13. Which VPN protocol provides **mobile device stability** and can auto-reconnect?

- A) PPTP
- B) IKEv2/IPSec
- C) GRE
- D) L2TP

Answer: B

Q14. Which two protocols are combined in L2TP/IPSec VPN?

- A) L2TP and TLS
- B) IPSec and HTTPS
- C) L2TP and IPSec
- D) PPTP and IPSec

Answer: C

Q15. What is the **primary security drawback** of PPTP?

- A) High CPU usage
- B) Uses TCP instead of UDP
- C) Weak encryption and vulnerable MS-CHAPv2
- D) Only available on Linux

Answer: C

Q16. Which VPN type connects **entire networks securely** over public infrastructure?

- A) SSL VPN
- B) Remote Access VPN
- C) Site-to-Site VPN
- D) IPsec Transport VPN

Answer: C

Q17. Secure VPNs implement which of the following mechanisms?

- A) Broadcast-based routing
- B) Address spoofing
- C) AES or ChaCha20 encryption
- D) Trust chaining only

Answer: C

Q18. What does "Perfect Forward Secrecy" ensure in VPNs?

- A) Permanent encryption keys
- B) IP spoofing is allowed
- C) Past sessions remain secure even if the key is compromised
- D) Public key reuse

Answer: C

Q19. Which VPN protocol is known for **fast setup, small codebase, and strong cryptography**?

- A) GRE
- B) OpenVPN
- C) WireGuard
- D) IKEv1

Answer: C

Q20. In VPN tunneling, which term best describes the **process of encapsulating data** inside another protocol?

- A) Encryption
- B) Routing
- C) Tunneling
- D) Packet marking

Answer: C

Q21. Trusted VPNs are **less secure** on the internet because:

- A) They use IPv4

- B) They lack encryption
- C) They require expensive hardware
- D) They operate in VLANs

Answer: B

Q22. What does the OpenVPN configuration file usually end in?

- A) .ovpn
- B) .vpn
- C) .openvpn
- D) .conf

Answer: A

Q23. In SSL VPN, which of the following is true?

- A) Needs a special VPN port
- B) Cannot bypass firewalls
- C) Can work over port 443 (HTTPS)
- D) Works only with IPsec

Answer: C

Q24. What encryption standard is widely used in **modern Secure VPNs**?

- A) DES
- B) 3DES

C) AES-256

D) RC4

Answer: C

Q25. A key benefit of site-to-site VPN is:

A) Anonymity for personal browsing

B) Connecting individual users to cloud

C) Secure permanent network links between office branches

D) Local DNS resolution

Answer: C

Q26. What is a potential **limitation** of SSL VPNs?

A) Incompatible with browsers

B) Cannot tunnel TCP

C) Limited access to network services (often only HTTP/S, FTP, RDP)

D) Requires a modem

Answer: C

Q27. IKEv2 uses which base protocol to manage keys?

A) TLS

B) HTTPS

C) ISAKMP

D) GRE

Answer: C

☒ *IKE (Internet Key Exchange) uses ISAKMP for negotiating and exchanging keys.*

Q28. In VPN terms, “split tunneling” means:

A) Creating multiple VPN servers

B) Routing some traffic via VPN and some directly to internet

C) Encryption is split between two protocols

D) Blocking IPv6

Answer: B

Q29. Which of these is **not** a VPN protocol?

A) WireGuard

B) OpenVPN

C) IPSec

D) SMB

Answer: D

Q30. In remote access VPNs, which authentication method is strongest?

A) Username only

B) Username + password

C) Multi-factor authentication (MFA)

D) IP address-based filtering

Answer: C

Q31. Which component in IPSec is responsible for negotiating and establishing security associations (SAs)?

A) Encapsulation Security Payload (ESP)

B) Authentication Header (AH)

C) IKE (Internet Key Exchange)

D) GRE (Generic Routing Encapsulation)

Answer: C

☒ *IKE negotiates cryptographic parameters and creates Security Associations (SAs) used by AH and ESP.*

Q32. Which transport layer protocol is primarily used by IPSec's ESP in tunnel mode?

A) TCP

B) UDP

C) Protocol number 50 (not TCP/UDP)

D) ICMP

Answer: C

☒ *ESP is not TCP/UDP; it is assigned IP protocol number 50.*

Q33. What's the **biggest challenge** with VPN passthrough on NAT devices for protocols like L2TP/IPSec?

A) Bandwidth limitations

B) Double encapsulation requires NAT-T (NAT Traversal)

C) DNS leakage

D) IP address conflicts

Answer: B

❑ *L2TP over IPSec requires NAT-T because multiple encapsulation layers don't work well with traditional NAT.*

Q34. Which encryption algorithm does **WireGuard** use for establishing secure connections?

- A) RSA-2048
- B) AES-CBC
- C) ChaCha20-Poly1305
- D) Blowfish

Answer: C

❑ *WireGuard uses modern, efficient ChaCha20 encryption and Poly1305 for authentication.*

Q35. Which statement best describes **Transport Mode** in IPSec?

- A) Entire IP packet is encrypted and encapsulated
- B) Only the IP payload is encrypted, headers remain visible
- C) It encapsulates TCP packets within IP headers
- D) Used only with trusted VPNs

Answer: B

Q36. In Secure VPN deployments, **Perfect Forward Secrecy (PFS)** ensures that:

- A) The same encryption key is reused for speed
- B) Sessions remain secure even if the private key is compromised
- C) Keys are never changed during a session
- D) VPN works only in remote access mode

Answer: B

☒ *PFS uses ephemeral keys so one session's compromise doesn't affect past/future sessions.*

Q37. Why is **OpenVPN** preferred over IPsec in dynamic IP or NAT environments?

- A) OpenVPN uses TCP by default
- B) It does not require certificates
- C) It is more lightweight
- D) It can operate over SSL/TLS on port 443, easily bypassing NAT/firewalls

Answer: D

Q38. A **Trusted VPN** setup in a telco environment typically relies on:

- A) Internet-based encrypted tunnels
- B) ISP routing and physical/logical separation over MPLS
- C) DNSSEC
- D) TLS handshake pinning

Answer: B

Q39. In OpenVPN, what purpose does `tls-auth` serve?

- A) Encrypt data packets
- B) Authenticate the user's password
- C) Prevent DoS and unauthenticated connection attempts
- D) Initiate NAT traversal

Answer: C

☒ *tls-auth adds an HMAC signature to control channel packets to filter out bogus requests.*

Q40. In a hybrid VPN deployment, which setup is most secure and scalable?

- A) Remote access VPN with public key encryption only
- B) Site-to-site VPN using static IP only
- C) SSL VPN for remote users + IPSec for inter-office WAN
- D) Trusted VPN over GRE tunnels

Answer: C

☒ *Combining SSL VPNs for flexible user access and IPSec for consistent, secure branch connectivity balances security and scalability.*